Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A hollow sphere is filled with water. It is hung by a long thread to make it a simple pendulum. As the water flows out of a hole at the bottom of the sphere, the frequency of oscillation will

(A) go on increasing

(B) go on decreasing

(C) first increases and then decreases

(D) first decreases and then increases

Q.2

A block is resting on a piston which executes simple harmonic motion in vertical plain with a period of $2.0\ \mathrm{s}$ in vertical plane at an amplitude just sufficient for the block to separate from the piston. The maximum velocity of the piston is

(A) $\frac{5}{\pi}\mathrm{m/s}$

(B) $\frac{10}{\pi}\mathrm{m/s}$

(C) $\frac{\pi}{2}\ \mathrm{m/s}$

(D) $\frac{20}{\pi}\mathrm{m/s}$

Q.3

The number of independent constituent simple harmonic motions yielding a resultant displacement equation of the periodic motion as $y = 8\sin^2(t/2)\sin(10t)$ is

(A) 8

(B) 6

(C) 4

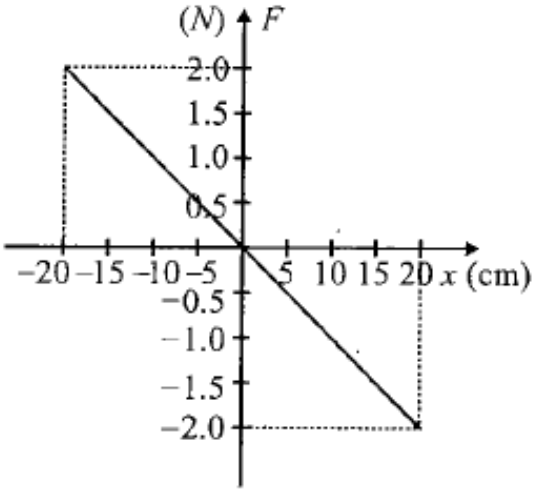
(D) 3

Q.4

Figure shows the variation of force acting on a particle of mass $400\ \mathrm{g}$ executing simple harmonic motion. The

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frequency of oscillation of the particle is



(A) $4\ \mathrm{s}^{-1}$

(B) $(5/2\pi)\mathrm{s}^{-1}$

(C) $(1/8\pi)\mathrm{s}^{-1}$

(D) $(1/2\pi)\mathrm{s}^{-1}$

Q.5

A particle executing SHM of amplitude 'a' has a displacement $a/2$ at $t = T/4$ and a negative velocity. The epoch of the particle is

(A) $\frac{\pi}{3}$

(B) $\frac{2\pi}{3}$

(C) π

(D) $\frac{5\pi}{3}$

Q.6

A particle executing harmonic motion is having velocities v_1 and v_2 at distances is x_1 and x_2 from the equilibrium position. The amplitude of the motion is

(A) $\sqrt{\frac{v_1^2x_2 - v_2^2x_1}{v_1^2 + v_2^2}}$

(B) $\sqrt{\frac{v_1^2x_1^2 - v_2^2x_2^2}{v_1^2 + v_2^2}}$

(C) $\sqrt{\frac{v_1^2x_2^2 - v_2^2x_1^2}{v_1^2 - v_2^2}}$

(D) $\sqrt{\frac{v_1^2x_2^2 + v_2^2x_1^2}{v_1^2 + v_2^2}}$

Q.7

Frequency of a particle executing SHM is $10\ \mathrm{Hz}$. The particle is suspended from a vertical spring. At the highest point of its oscillation the spring is unstretched Maximum speed of the particle is $(g = 10\ \mathrm{m/s^2})$

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(A) $2\pi \mathrm{m/s}$

(B) $\pi \mathrm{m/s}$

(C) $1/\pi \mathrm{m/s}$

(D) $1/2\pi \mathrm{m/s}$

Q.8

The potential energy of a particle of mass $1\ \mathrm{kg}$.in motion along the x-axis is given by: $U = 4(1 - \cos 2x)$, where x is in metres. The period of small oscillation (in seconds) is

(A) 2π

(B) π

(C) $\pi/2$

(D) $\sqrt{2}\pi$

Q.9

The K.E. and P.E. of a particle executing SHM with amplitude A will be equal when its displacement is:

(A) $A\sqrt{2}$

(B) $A/2$

(C) $A/\sqrt{2}$

(D) $A\sqrt{2/3}$

Q.10

Two particles move parallel to the x-axis about the origin with same amplitude ' a ' and frequency ω. At a certain instant they are found at a distance $a/3$ from the origin on opposite sides but their velocities are in the same direction. What is the phase difference between the two?

(A) $\cos^{-1}\frac{7}{9}$

(B) $\cos^{-1}\frac{5}{9}$

(C) $\cos^{-1}\frac{4}{9}$

(D) $\cos^{-1}\frac{1}{9}$

Q.11

The potential energy of a particle executing SHM along the x-axis is given by $U = U_0 - U_0 \cos ax$. What is the period of oscillation?

(A) $2\pi\sqrt{\frac{ma}{U_0}}$

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(B) $2\pi\sqrt{\frac{U_0}{ma}}$

(C) $\frac{2\pi}{a}\sqrt{\frac{m}{U_0}}$

(D) $2\pi\sqrt{\frac{m}{aU_0}}$

Q.12

A simple pendulum of length l and a mass m of the bob is suspended in a car that is travelling with a constant speed v around a circle of radius R. If the pendulum undergoes small oscillations about its equilibrium position, the frequency of its oscillation will be

(A) $\frac{1}{2\pi}\sqrt{\frac{g}{l}}$

(B) $\frac{1}{2\pi}\sqrt{\frac{g}{R}}$

(C) $\frac{1}{2\pi}\sqrt{\frac{\left(g^2+\frac{v^4}{R^2}\right)^{1/2}}{l}}$

(D) $\frac{1}{2\pi}\sqrt{\frac{v^2}{Rl}}$

Q.13

In a certain oscillatory system (particle is performing SHM), the amplitude of motion is $5\ \mathrm{m}$ and the time period is $4\ \mathrm{s}$. The minimum time taken by the particle for passing between points, which are at distances of $4\ \mathrm{m}$ and $3\ \mathrm{m}$ from the centre and on the same side of it will approximately be

(A) $\frac{16}{45}\ \mathrm{s}$

(B) $\frac{7}{45}\ \mathrm{s}$

(C) $\frac{8}{45}\ \mathrm{s}$

(D) $\frac{13}{45}\ \mathrm{s}$

Q.14

A particle executes SHM with time period $8\ \mathrm{s}$. Initially, it is at its mean position. The ratio of distance travelled by it in the 1st second to that in the 2nd second is

(A) $\sqrt{2}:1$

(B) $1:(\sqrt{2}-1)$

(C) $(\sqrt{2}+1):\sqrt{2}$

(D) $(\sqrt{2}-1):1$

Q.15

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The instantaneous displacement x of a particle executing simple harmonic motion is given by $x = a_1 \sin \omega t + a_2 \cos(\omega t + \pi/6)$. The amplitude A of oscillation is given by

(A) $\sqrt{a_1^2 + a_2^2 + 2a_1 a_2 \cos \frac{\pi}{6}}$

(B) $\sqrt{a_1^2 + a_2^2 + 2a_1 a_2 \cos \frac{\pi}{3}}$

(C) $\sqrt{a_1^2 + a_2^2 - 2a_1 a_2 \cos \frac{\pi}{6}}$

(D) $\sqrt{a_1^2 + a_2^2 - 2a_1 a_2 \cos \frac{\pi}{3}}$

Q.16

A particle performs SHM about $x = 0$ such that at $t = 0$ it is at $x = 0$ and moving towards positive extreme. The time taken by it to go from $x = 0$ to $x = A/2$ is .... times the time taken to go from $x = A/2$ to A. The most suitable option for the blank space is

(A) 2

(B) $\frac{1}{2}$

(C) $\frac{11}{12}$

(D) $\frac{12}{11}$

Q.17

A particle performs simple harmonic motion with amplitude A and time period T. The mean velocity of the particle over the time interval during which it travels a distance of $A/2$ starting from extreme position is

(A) $\frac{A}{T}$

(B) $\frac{2A}{T}$

(C) $\frac{3A}{T}$

(D) $\frac{A}{2T}$

Q.18

Two springs are made to oscillate simple harmonically due to the same mass individually. The time periods obtained are T_1 and T_2. If both the springs are connected in series and then made to oscillate by the same mass, the resulting time period will be

(A) $T_1 + T_2$

(B) $\frac{T_1 T_2}{T_1 + T_2}$

(C) $\sqrt{T_1^2 + T_2^2}$

(D) $\frac{T_1 + T_2}{2}$

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Q.19

A particle is performing SHM according to the equation $x = (3\text{ cm})\sin\left(\frac{2\pi}{18}t + \frac{\pi}{6}\right)$, where t is in seconds. The distance travelled by the particle in 39 s is

(A) 24 cm

(B) 1.5 cm

(C) 25.5 cm

(D) None of these

Q.20

A particle performs SHM on the x-axis with amplitude A and time period T. The time taken by the particle to travel a distance $A/5$ starting from rest is

(A) $\frac{T}{20}$

(B) $\frac{T}{2\pi}\cos^{-1}\left(\frac{4}{5}\right)$

(C) $\frac{T}{2\pi}\cos^{-1}\left(\frac{1}{5}\right)$

(D) $\frac{T}{2\pi}\sin^{-1}\left(\frac{1}{5}\right)$

Q.21

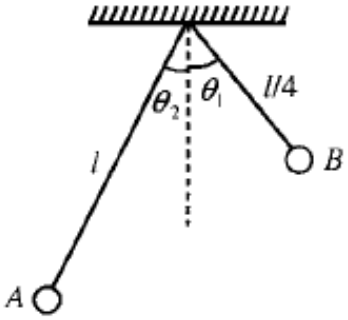
A particle of mass 0.50 kg executes a simple harmonic motion under a force $F = -(\mathbf{50\ N/m})x$. If it crosses the centre of oscillation with a speed of 10 m/s, find the amplitude of motion.

Q.22

If the displacement of a moving point at any time is given by an equation of the form $y(t) = a\cos\omega t + b\sin\omega t$, show that the motion is simple harmonic. If $a = 3$ m, $b = 4$ m and $\omega = 2$. Determine the maximum velocity in m/sec.

Q.23

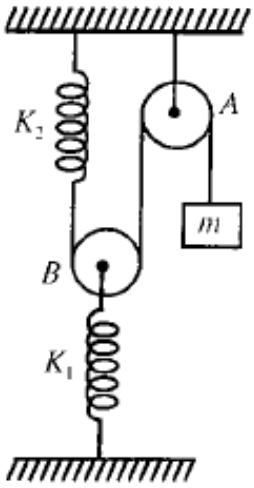
Two simple pendulums A and B having lengths l and $l/4$ respectively are released from the position as shown in Fig. Calculate the time (in seconds) after which the two strings become parallel for the first time. (Take $\ell = \frac{90}{\pi^2}m$ and $g = 10\text{ m/s}^2$)



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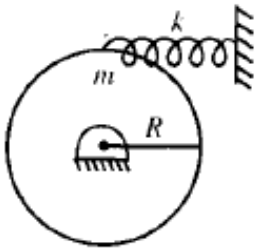
Q.24

A block of mass m is tied to one end of a spring which passes over a smooth fixed pulley A and under a light smooth movable pulley B. The other end of the string is attached to the lower end of a spring of spring constant K_2. Find the period of small oscillation of mass m about its equilibrium position (in second). (Take $m = \pi^2$ kg, $K_2 = 4K_1$ and $K_1 = 17$ N/m.)



Q.25

A uniform disc of mass m and radius R is pivoted smoothly at its centre of mass. A light spring of stiffness k is attached with the disc tangentially as shown in the Fig. Find the angular frequency in rad/s of torsional oscillations of the disc. (Take $m = 5$ kg and $K = 10$ N/m.)



Answers & Solutions

Q.1 Answer:

first decreases and then increases

Solution:

Frequency of a simple pendulum

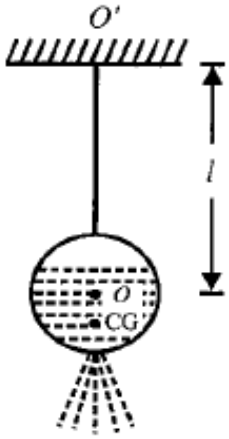
$f = \frac{1}{2\pi}\sqrt{\frac{g}{l}}$

Evidently, $f \propto \frac{1}{\sqrt{l}}$ at any place.

The effective length l is the distance of the point of suspension O' from the centre of gravity (CG) of the bob. As, water flows out of the hole at the bottom, the CG descends from centre towards the

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bottom, increasing the effective length, and consequently f decreases.



However, when all the water has flows out, the CG of a hollow sphere is once again at its centre and hence the effective length would decrease, thereby increasing the frequency.

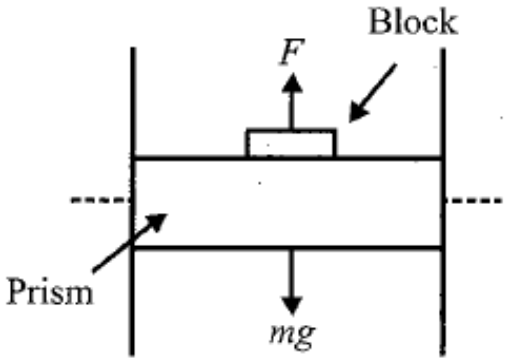
Q.2 Answer:

$\frac{10}{\pi}\mathrm{m/s}$

Solution:

The situation when the block is just below the mean position is shown in Fig., the restoring forces acting on the piston cause a normal reaction F to act on the block. For the block to separate $F \geq mg$ i.e., $m\omega^2 A \geq mg$

(where $\omega =$ angular frequency and $A =$ amplitude)



or $A \geq \frac{g}{\omega^2} = \frac{g}{\left(\frac{2\pi}{T}\right)^2}$ **or** $A \geq \frac{gT^2}{4\pi^2}$

Now, the maximum velocity $v_{\max}$ at that instant $= \omega A$

$v_{\max} = \left(\frac{2\pi}{T}\right)\left(\frac{gT^2}{4\pi^2}\right) = \frac{10}{\pi}\mathrm{m/s^2}$

Q.3 Answer:

3

Solution:

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$$y = 8\sin^2\left(\frac{t}{2}\right)\sin(10t)$$

$$= 4[1-\cos t]\sin(10t)\ \left(\text{using } 2\sin^2\frac{\theta}{2} = 1-\cos\theta\right)$$

$$= 4\sin(10t) - 4\sin(10t)\cos t$$

$$= 4\sin(10t) - 2[\sin 11t + \sin 9t]$$

$$\left(\text{using } 2\sin C\cos D = \sin(C+D) + \sin(C-D)\right)$$

$$= 4\sin(10t) - 2\sin(11t) - 2\sin(9t)$$

Evidently, y is obtained as the superimposition of three independent (i.e., having different angular frequency ω) SHMs.

Q.4 Answer:

$(5/2\pi)\mathrm{s}^{-1}$

Solution:

The slope of the length

$= \frac{F}{x} = -\frac{0.5}{5} = -0.1\ \mathrm{N/cm} = -10\ \mathrm{N/m}$

But $F = -m\omega^2 x$ **or,** $F/x = -m\omega^2$

so, $-m\omega^2 = -10$ **or,** $m\omega^2 = 10$

$\text{or, } \omega^2 = 10/\mathrm{m}$

$\therefore \quad \omega^2 = \frac{10}{4\times 10^{-1}} \Rightarrow \omega = \frac{10}{2} = 5$

$\therefore \quad f = \frac{\omega}{2\pi} = \frac{5}{2\pi}/\mathrm{s}$

Q.5 Answer:

$\frac{\pi}{3}$

Solution:

Equation of $\mathrm{SHM} = y\sin(\omega t + \phi)$

When $y = \frac{a}{2}, t = \frac{T}{4} = \frac{2\pi}{4\omega} = \frac{\pi}{2\omega}$

$v = a\omega\cos(\omega t + \phi)$**, velocity is negative**

$\frac{\pi}{2} < (\omega t + \phi) < \frac{3\pi}{2}$

$\frac{a}{2} = a\sin(\omega t + \phi) \quad \Rightarrow \sin(\omega t + \phi) = \frac{1}{2}$

$\left(\frac{\pi}{2} + \phi\right) = \frac{5\pi}{6}$

Substituting in the above equation, we get $\phi = \pi/3$

Q.6 Answer:

$\sqrt{\frac{v_1^2x_2^2 - v_2^2x_1^2}{v_1^2 - v_2^2}}$

Solution:

$$v_1 = \omega\sqrt{a^2 - x_1^2}, \quad v_2 = \omega\sqrt{a^2 - x_2^2}$$

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We get, $a = \sqrt{\frac{v_1^2 x_2^2 - v_2^2 x_1^2}{v_1^2 - v_2^2}}$

Q.7 Answer:

$1/2\pi \mathrm{m/s}$

Solution:

Mean position of the particle is mg/k **distance below the unstretched position of spring. Therefore, amplitude of oscillation is** $A = \frac{mg}{k}$

$\omega = \sqrt{\frac{k}{m}} = 2\pi f = 20\pi (f = 10 \text{ Hz})$

$\frac{m}{k} = \frac{1}{400\pi^2}$

$v_{\max} = A\omega = \frac{g}{400\pi^2} \times 20\pi = \frac{1}{2\pi} \mathrm{m/s}$

Q.8 Answer:

$\pi/2$

Solution:

$F = -\frac{dU}{dx} = -8 \sin 2x$

For small oscillations, $\sin 2x = 2x$ **i.e.,**

$a = -16x$

Since $a\mu - x$**, the oscillations are simple harmonic in nature.**

$T = 2\pi \sqrt{|\ \frac{x}{a}\ |} = 2\pi \sqrt{\frac{1}{16}} = \frac{\pi}{2} \text{ s}$

Q.9 Answer:

$A/\sqrt{2}$

Solution:

$K = \frac{1}{2} m\omega^2 \left(A^2 - y^2\right)$

$U = \frac{1}{2} m\omega^2 y^2$

$K + U = \frac{1}{2} m\omega^2 \left(A^2 - y^2\right) + \frac{1}{2} m\omega^2 y^2$

i.e., $2y^2 = A^2$ **or** $y = \frac{A}{\sqrt{2}}$

Q.10 Answer:

$\cos^{-1} \frac{7}{9}$

Solution:

Let $x_1 = a \sin \omega t$ **and** $\dot{x}_2 = a \sin(\omega t + \delta)$ **be two SHMs.**

$\frac{a}{3} = a \sin \omega t$ and $-\frac{a}{3} = a \sin(\omega t + \delta)$

$\sin \omega t = 1/3$ **and** $\sin(\omega t + \delta) = -1/3$

Eliminating t, $\frac{1}{3} \cos \delta + \sqrt{1 - \frac{1}{9}} \sin \delta = -\frac{1}{3}$

$9 \cos^2 \delta + 2 \cos \delta - 7 = 0$

$\cos \delta = -1$ or $\frac{7}{9}$

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i.e., $\delta = 180^\circ$ or $\cos^{-1}\left(\frac{7}{9}\right)$

If we put 180°, we find that v_1 and v_2 are of opposite signs. Hence $\delta = 180^\circ$ is not applicable.

$\therefore \quad \delta = \cos^{-1}\left(\frac{7}{9}\right)$

Q.11 Answer:

$\frac{2\pi}{a}\sqrt{\frac{m}{U_0}}$

Solution:

$$U = U_0 - U_0 \cos ax$$

$$F = -\frac{dU}{dx} = -U_0 \sin ax = m\frac{d^L x}{dt^2}$$

This is the equation of SHM $\Rightarrow T = \frac{2\pi}{a}\sqrt{\frac{m}{U_0}}$

Q.12 Answer:

$\frac{1}{2\pi}\sqrt{\frac{\left(g^2+\frac{v^4}{R^2}\right)^{1/2}}{l}}$

Solution:

The centripetal acceleration on the bob as it oscillates (acting along the radius of circle) $= v^2/R$.

This will act horizontally towards the centre of circular path.

The total acceleration acting on the pendulum bob is therefore $a = \sqrt{g^2 + \left(\frac{v^2}{R}\right)^2}$

The frequency of oscillation will therefore be

$n = \frac{1}{2\pi}\sqrt{\frac{a}{l}} = \frac{1}{2\pi}\sqrt{\frac{\left(g^2+\frac{v^4}{R^2}\right)^{1/2}}{l}}$

Q.13 Answer:

$\frac{8}{45}$ s

Solution:

$x = A\sin\omega t$, **where** $\omega = \frac{2\pi}{4} = \frac{\pi}{2}\text{rad/s}$

$3 = 5\sin\omega t_1 \Rightarrow t_1 = \frac{1}{\omega}\sin^{-1}\left(\frac{3}{5}\right) = \frac{2\times 37}{180}$

$4 = 5\sin\omega t_2 \Rightarrow t_2 = \frac{1}{\omega}\sin^{-1}\left(\frac{4}{5}\right) = \frac{2\times 53}{180}$

Hence time interval $(t_2 - t_1) = \frac{1}{90}(53 - 37) = \frac{8}{45}$ s

Q.14 Answer:

$1 : (\sqrt{2} - 1)$

Solution:

$x = a\sin\omega t$ **with** $\omega = \frac{2\pi}{T} = \frac{2\pi}{8} = \frac{\pi}{4}\text{rad/s}$

$x = a\sin\frac{\pi t}{4}$

In 2 s (which is equal to $T/4$), one amplitude will be covered. In lst second

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$x = a \sin \frac{\pi}{4} = \frac{a}{\sqrt{2}}$

Required ratio $= -\frac{\frac{a}{\sqrt{2}}}{a - \frac{a}{\sqrt{2}}} = \frac{1}{\sqrt{2}-1}$

Q.15 Answer:

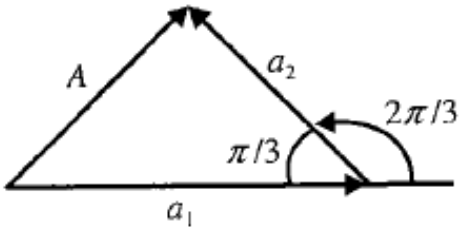
$\sqrt{a_1^2 + a_2^2 - 2a_1 a_2 \cos \frac{\pi}{3}}$

Solution:

$$x = a_1 \sin \omega t + a_2 \cos\left(\omega t + \frac{\pi}{6}\right)$$

$$x = a_1 \sin \omega t + a_2 \sin\left(\omega t + \frac{\pi}{6} + \frac{\pi}{2}\right)$$

$$\text{or, } x = a_1 \sin \omega t + a_2 \sin\left(\omega t + \frac{2\pi}{3}\right)$$



$$\therefore \quad A^2 = a_1^2 + a_2^2 + 2a_1 a_2 \cos \frac{2\pi}{3}$$

$$= a_1^2 + a_2^2 - 2a_1 a_2 \cos \frac{\pi}{3}$$

$$\therefore \quad A = \sqrt{a_1^2 + a_2^2 - 2a_1 a_2 \cos \frac{\pi}{3}}$$

Q.16 Answer:

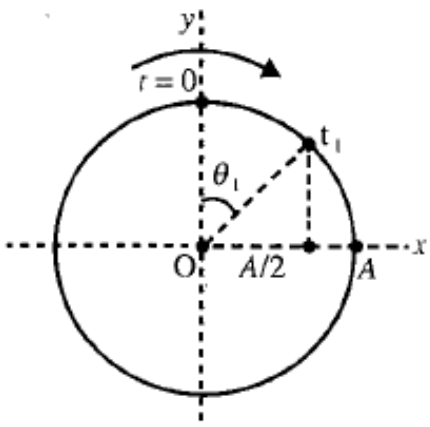
$\frac{1}{2}$

Solution:

From circular motion representation, we can represent SHM by uniform circular motion.

Let t_1 is the time taken by particle to go from $x = 0$ to $x = A/2$

$\sin \theta_1 = \frac{A/2}{A} = \frac{1}{2}$



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$\theta_1 = \frac{\pi}{6} \quad \Rightarrow \omega t_1 = \frac{\pi}{6} \quad \Rightarrow t_1 = \frac{\pi \times T}{6 \times 2\pi} = \frac{T}{12}$

So, time taken to go from $x = A/2$ to A is

$t_2 = \frac{T}{4} - t_1 = \frac{T}{4} - \frac{T}{12} = \frac{T}{6}$

Hence, $\frac{t_1}{t_2} = \frac{T/12}{T/6} = \frac{1}{2}$

Q.17 Answer:

$\frac{3A}{T}$

Solution:

Time interval is $T/6$

Let the equation of simple harmonic motion be $x = A\cos\omega t, v = -A\sin\omega t$**, where** $\omega = 2\pi/T$

Mean velocity over the required interval is

$\langle v \rangle = \frac{\int_0^{T/6} v dt}{T/6}$

$= \frac{\int_0^{T/6} -A\omega \sin\omega t dt}{T/6} = \frac{6A}{T}\cos\omega t\Big|_0^{T/6} = \frac{-3A}{T}$

Q.18 Answer:

$\sqrt{T_1^2 + T_2^2}$

Solution:

Let spring constant of two springs are k_1 **and** k_2**, respectively, then**

$T_1 = 2\pi\sqrt{\frac{m}{k_1}} \quad \text{and} \quad T_2 = 2\pi\sqrt{\frac{m}{k_2}}$

$k_1 = \frac{4\pi^2 m}{T_1^2} \quad \text{and} \quad k_2 = \frac{4\pi^2 m}{T_2^2}$

When the two springs are connected in series, then $T = 2\pi\sqrt{\frac{m}{k_{eq}}}$ **where** $k_{\text{eq}} = \frac{k_1 k_2}{k_1 + k_2}$

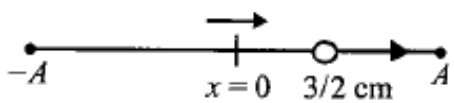
$T = \sqrt{T_1^2 + T_2^2}$

Q.19 Answer:

25.5 cm

Solution:

The time period of simple harmonic motion is $T = 18$ s.



At $t = 0$**, the particle is at** $x = 3/2$ cm **and approaching positive extreme.**

$39 \text{ s} = 2 \times 18 + 3 = 2T + 3 = 2T + T/6$

Distance travelled by the particle in $2T$ **is** $8A = 24$ cm**. Distance travelled by the particle in**

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further $T/6$ is $A/2 = 1.5$ cm.

Total distance travelled $= 25.5$ cm

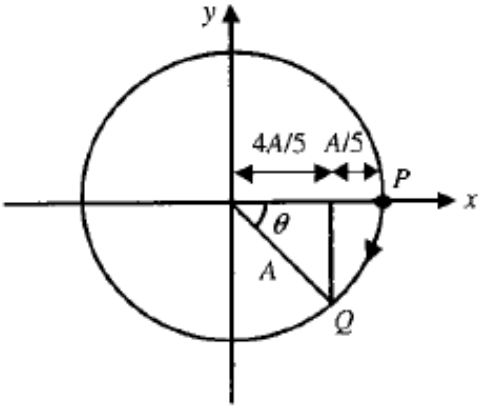
Q.20 Answer:

$\frac{T}{2\pi}\cos^{-1}\left(\frac{4}{5}\right)$

Solution:

Method 1: Particle is starting from rest, i.e., from one of its extreme position.

As particle moves a distance $A/5$, we can represent it on a circle as shown.



$\cos\theta = \frac{4A/5}{A} = \frac{4}{5} \quad \Rightarrow \quad \theta = \cos^{-1}\left(\frac{4}{5}\right)$

$\omega t = \cos^{-1}\left(\frac{4}{5}\right) \Rightarrow t = \frac{1}{\omega}\cos^{-1}\left(\frac{4}{5}\right) = \frac{T}{2\pi}\cos^{-1}\left(\frac{4}{5}\right)$

Method 2: As the particle starts from rest, i.e., from extreme position $x = A\sin(\omega t - \phi)$

At $t = 0; \quad x = A \quad \Rightarrow \phi = \frac{\pi}{2}$

$A - \frac{A}{5} = A\cos\omega t$

$\frac{4}{5} = \cos\omega t \Rightarrow \omega t = \cos^{-1}\frac{4}{5}$

$t = \frac{T}{2\pi}\cos^{-1}\left(\frac{4}{5}\right)$

Q.21 Answer:

—

Solution:

The kinetic energy of the particle when it is mean position, it is also its total energy as the potential energy is zero here.

$E = \frac{1}{2}mv^2 = \frac{1}{2}(0.50\text{ kg})(10\text{ m/s})^2 = 25\text{ J}$

At the maximum displacement $x = A$, the speed is zero and hence the kinetic energy is zero. The potential energy here is $1/2kA^2$. At this position it will also its total energy. Hence

$\frac{1}{2}kA^2 = 25\text{ J}$ **..(i)**

The force on the particle is given by, $F = -(50\text{ N/m})x$ **Thus, the spring constant is** $k = 50\text{ N/m}$.

Equation (i) gives

$\frac{1}{2}(50\text{ N/m})A^2 = 25\text{ J} \Rightarrow A = 1\text{ m}$

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Q.22 Answer:

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Solution:

The particle is moving along the y-axis.

$$y(t) = a\cos\omega t + b\sin\omega t$$

$$y = \sqrt{a^2+b^2}\sin(\omega t + \phi_0)$$

where $\tan\phi_0 = a/b \Rightarrow y = \sqrt{a^2+b^2}\sin\left(\omega t + \tan^{-1}a/b\right)$

Comparing with $y = a\sin(\omega t + \phi)$,

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{2} = \pi\text{s}$$

$$A = \sqrt{a^2+b^2} = \sqrt{3^2+4^2} = 5\text{ m}$$

$$v_{\max} = \omega A = 2\times 5 = 10\text{ m/s}$$

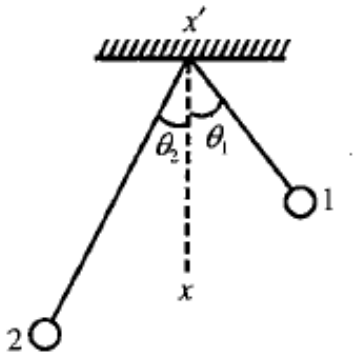
$$\Rightarrow \quad x_{\max} = \omega^2 A = 4\times 5 = 20\text{ m/s}^2$$

Q.23 Answer:

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Solution:

The angular positions of pendulums 1 and 2 are (taking angles to the right of reference line xx' to be positive)



$$\theta_1 = \theta\cos\left(\frac{4\pi}{T}t\right) \quad \left(\text{where } T = 2\pi\sqrt{\frac{l}{g}}\right)$$

$$\theta_2 = -\theta\cos\left(\frac{2\pi}{T}t\right) = \cos\left(\frac{2\pi}{T}t + \pi\right)$$

For the strings to be parallel for the first time

$$\theta_1 = \theta_2$$

$$\cos\left(\frac{4\pi}{T}t\right) = \cos\left(\frac{2\pi}{T}t + \pi\right)$$

$$\therefore \frac{4\pi}{T}t = 2n\pi \pm \left(\frac{2\pi}{T}t + \pi\right)$$

$$\text{for } n = 0, t = T/2$$

$$\text{for } n = 1, t = T/6, 3T/2$$

Both the pendulums are parallel to each other for the first time after

$$t = \frac{T}{6} = \frac{\pi}{3}\sqrt{\frac{l}{g}} = 1\text{ s}$$

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Q.24 Answer:

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Solution:

Net increment in the tension in string connecting the block will provide acceleration to the block. We can write

$\Delta T = ma$

When the block m is displaced by a distnce x beyond equilibrium position, the addition stretch of springs 1 and 2 are x_1 and x_2 respectively.

We can write

$x_1 = \frac{x - x_2}{2} \Rightarrow x = 2x_1 + x_2$ **...(ii)**

As $\Delta T = k_2 x_2$ **and** $2\Delta T = k_1 x_1$ **then** $2\left(k_2 x_2\right) = k_1 x_1$

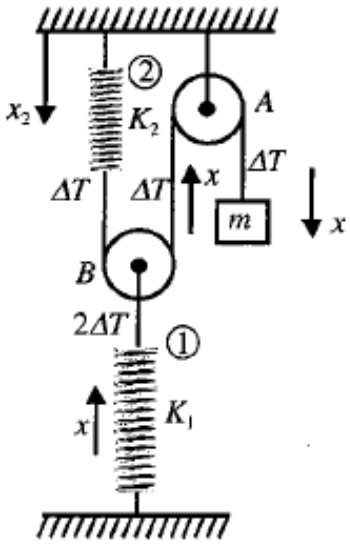
$\Rightarrow \quad x_1 = \frac{2k_2 x_2}{k_1}$ **...(iii)**

from Eqs. (ii) and (iii) $x = 2\left[\frac{2k_2 x_2}{k_1}\right] + x_2 = \frac{(4k_2 + k_1)}{k_1} x_2 \Rightarrow \quad x_2 = \frac{k_1 x}{(k_1 + 4k_2)}$

and $\vec{a} = \frac{-k_1 k_2}{(k_1 + 4k_2)m} \cdot x \quad \Rightarrow \quad \vec{a} = \frac{-k_1 k_2}{(k_1 + 4k_2)m} \cdot x\,\omega^2 = \frac{k_1 k_2}{(k_1 + 4k_2)m}$

Hence $R = \frac{80}{23\pi^2} m$

After substituting the values, we get $T = 1$ **s.**



Q.25 Answer:

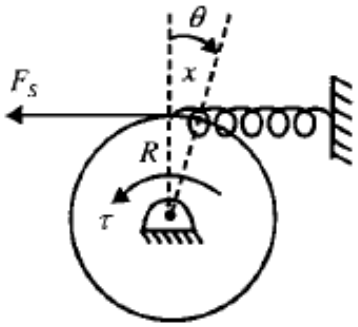
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Solution:

If we twist (rotate) the disc through a small clockwise angle θ, the spring will be deformed (compressed) by a distance $x = R\theta$. Hence, the spring force $F_s = kx = k(R\theta)$ will produce a

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restoring torque.



Restoring torque: $\tau = -F_S R$ **where** $F_S = kR\theta$

This gives $\tau = -kR^2\theta$

It means after removing the external (applied) torque, the restoring torque rotates the disc with an angular acceleration α **which will bring the spring-disc system back to its original state.**

Newton's law of rotation (or torque equation): Applying Newton's second law of rotation, we have

$\tau = I_C\alpha$

where $\tau = -kR^2\theta$

This gives $\alpha = -\frac{kR^2\theta}{I_C}$ **where** $I_C = \frac{mR^2}{2}$

Then $a = -\frac{2k}{m}\theta$

Comparing the above equation with $\alpha = -\omega^2\theta$**, we have**

$\omega = \sqrt{\frac{2k}{m}}$

After substituting the values we get $\omega = 2\text{rad/s}$.

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